

INTRO TO WEB DEVELOPMENT

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Unit 1 - The Web and Web Development

Part 1: Screenshot of Work

Welcome to my intro page



Sample picture

About me

Hello, my name is **Yuantao Huang**, I am

I am currently learning editing html file

My skill

- cpp
- cpp
- cpp

Learning process

1. iostream library
2. string library
3. vector library

Contact me

E-mail:

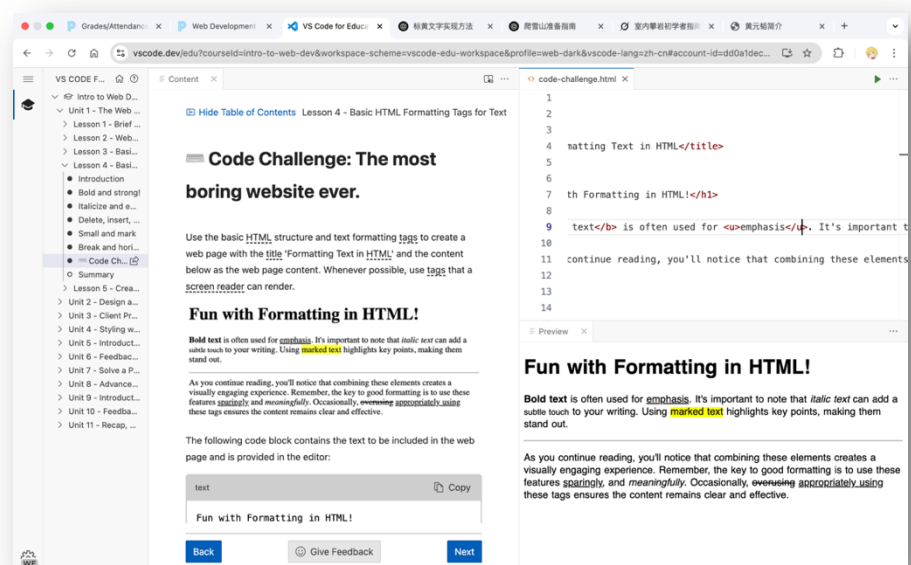
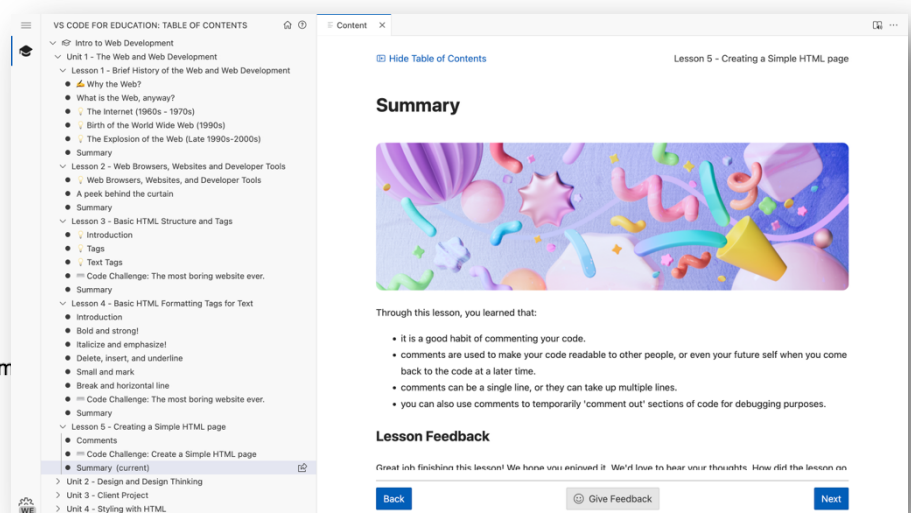
ylmpdy@126.com

Github:

[yypdylm](https://github.com/yypdylm)

Sample table

name	age	job	location
Tom	28	engineer	Beijin
Jerry	32	engineer	Shanghai
James	25	engineer	Shenzhen



Part 2: Code Snippet



index1.html

Uploaded in the zip file: unit_1.zip

Part 3: Reflection

In Unit 1, I learned the basics of HTML and how to use tags to structure a webpage. For example, I used the `<b>` tag to make my name bold, emphasizing it on the page. The `<u>` tag helped me underline the word "html" to highlight what I'm learning. I also applied the `<i>` tag to italicize "high school student," making the text style different, and the `<mark>` tag to highlight my name in yellow, drawing attention to it. Additionally, I used heading tags like `<h1>`, `<h2>`, and `<h3>` to organize my content into sections, and the `<p>` tag to write paragraphs. I also learned how to insert an image using the `<img>` tag, setting its width and height to fit the page nicely. These tags helped me understand how HTML controls the appearance and structure of a webpage.

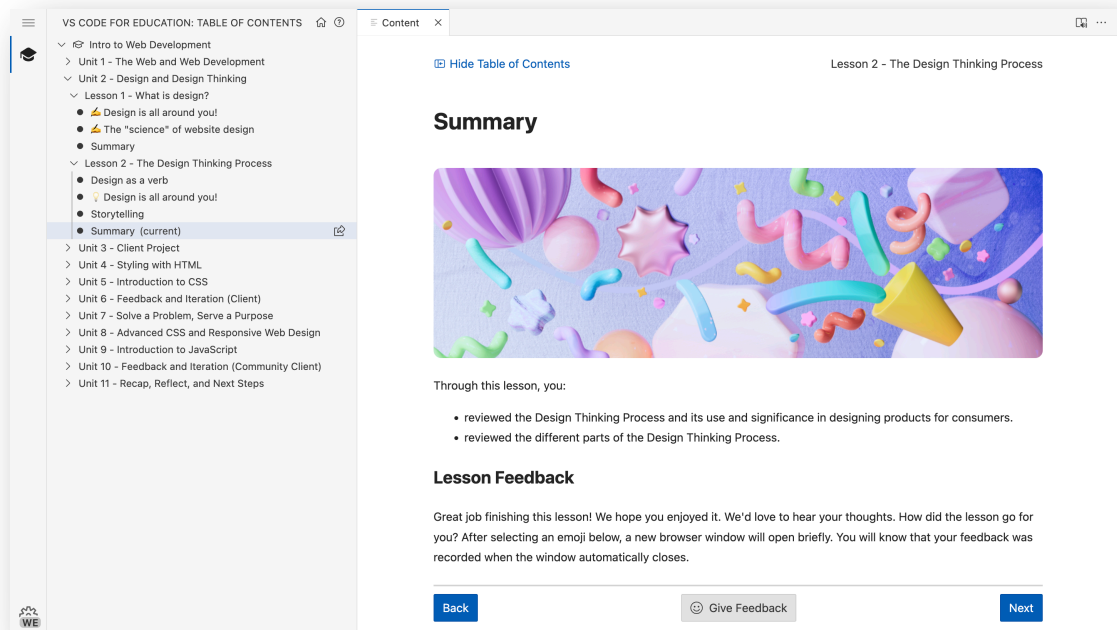
At first, I struggled to remember the purpose of each tag, like confusing `<b>` for bold with `<i>` for italics, and I wasn't sure how to properly nest tags like `<b>` and `<mark>` together.

I practiced by writing and testing the code multiple times in VS Code, checking the browser output to see how each tag affected the page. I also reviewed the lesson notes to clarify the differences between tags and practiced nesting them correctly.

This unit taught me the foundational role of HTML in web development. By learning these basic tags, I can now structure content on a webpage, which is the first step before adding styles with CSS or functionality with JavaScript in future lessons.

Unit 2 - Design and Design Thinking

Part 1: Screenshot of Work



Part 2: Code Snippet



index2.html

Uploaded in the zip file: unit_2.zip

Part 3: Reflection

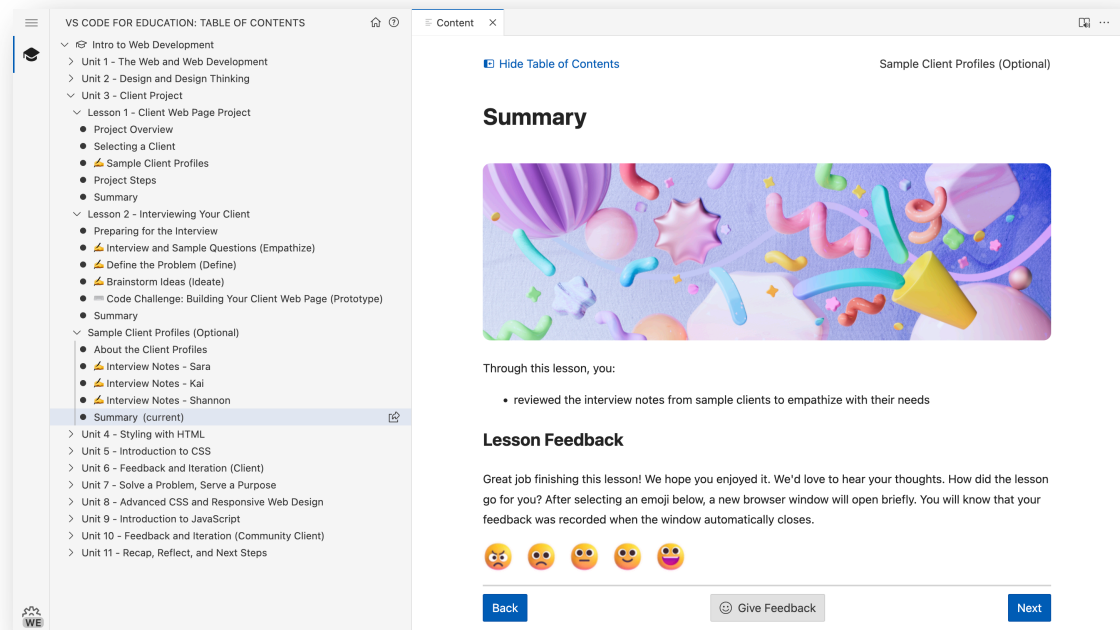
In Unit 2, I learned about design principles and the Design Thinking Process, which help make websites better for users. Lesson 1 taught me how functionality, aesthetics, and user experience (UX) work together in product design. I practiced looking at websites and thinking about their user interface (UI) and UX, like how easy they are to use. Lesson 2 showed me the Design Thinking Process, with steps like understanding users and testing ideas, which I can use to plan a webpage.

At first, I struggled to understand these big ideas because they weren't about coding like Unit 1. I wasn't sure how design connects to making a website without using HTML right away. To fix this, I read the lesson summaries again and wrote notes about UI and UX examples. I also tried sketching a webpage layout with sections like a header and content area to see how design works.

This unit ties into web development by showing me how to think about users before coding. It builds on Unit 1's HTML structure and prepares me for CSS styling in Unit 5 and JavaScript in Unit 9, making sure my webpages are both useful and nice to look at.

Unit 3 – Client Project

Part 1: Screenshot of Work



Part 2: Code Snippet



index3.html

Uploaded in the zip file: unit_3.zip

Part 3: Reflection

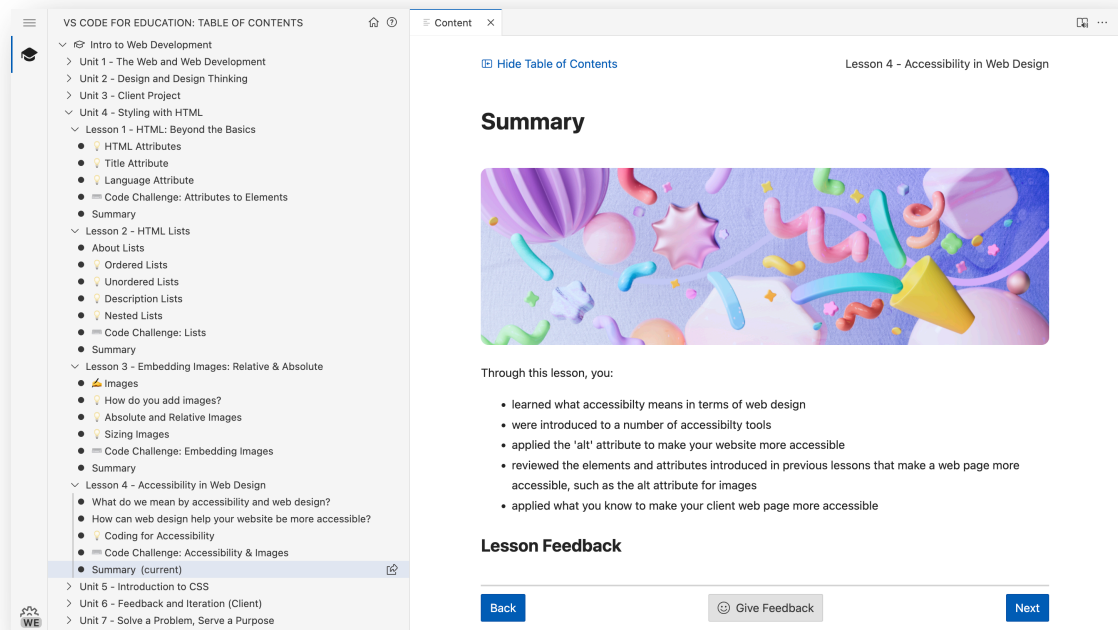
In Unit 3, I learned how to apply the Design Thinking Process to design a web page for a client. I chose Shannon as my client in Lesson 1 because her requirements were clear—she wants a tech-focused website combining Instagram and Spotify features, inspired by her passion for innovation.

In Lesson 2, interviewing her helped me empathize with her needs, and I created a 'define' statement to focus my design. Brainstorming ideas was tricky since her concept was ambitious, but I overcame this by sketching layouts and prioritizing key features like music and visuals. Building the prototype in HTML was exciting, as it let me apply coding skills to meet her vision.

The biggest challenge was balancing creativity with simplicity, which I managed by starting with a basic structure. This unit taught me the value of understanding a client's needs before coding, a key part of web development. It connects to the broader process by showing how user-centered design drives effective websites.

Unit 4 – Styling with HTML

Part 1: Screenshot of Work



Part 2: Code Snippet



index4.html

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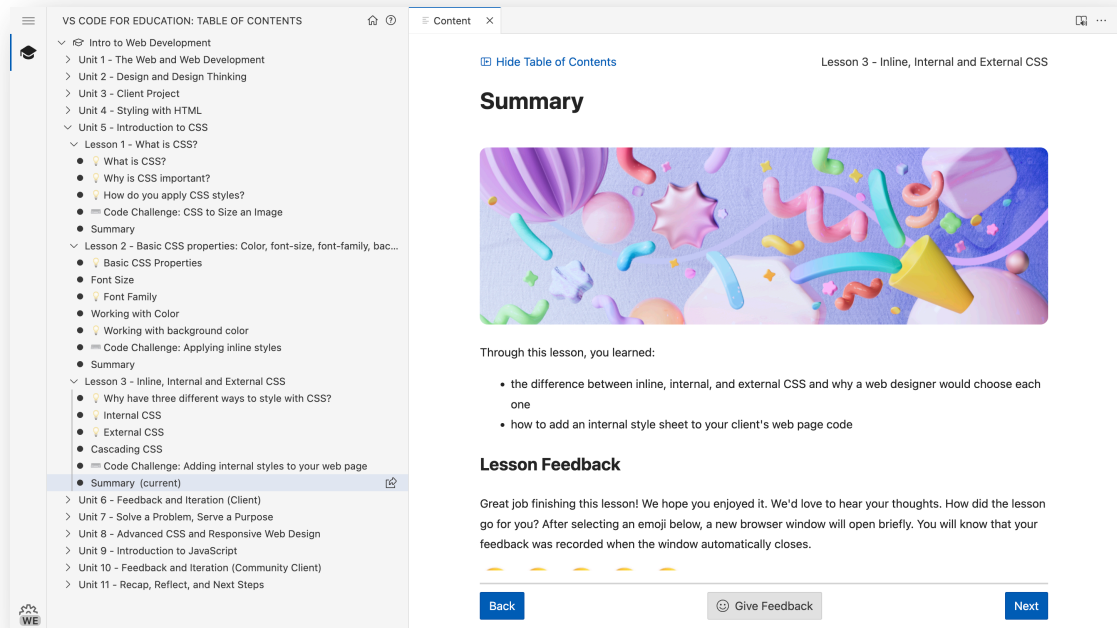
Part 3: Reflection

In Unit 4, I gained essential skills for enhancing web pages with HTML. Lesson 1 taught me how to format attributes correctly, use the title attribute to add information, and apply the lang attribute to specify the page's language. Lesson 2 showed me how to code ordered, unordered, and description lists, and adjust their presentation with attributes. In Lesson 3, I learned to embed images with attributes like alt for accessibility, size them properly, and understand relative versus absolute links. Lesson 4 emphasized accessibility, introducing tools and reinforcing the alt attribute's role in inclusive design.

Challenges included memorizing attribute syntax and writing effective alt text. I overcame these by practicing with examples and using HTML validators to check my code. This unit connects to web development by laying the foundation for creating structured, accessible, and visually engaging pages—skills I'll build on with CSS and JavaScript later.

Unit 5 – Introduction to CSS

Part 1: Screenshot of Work



Part 2: Code Snippet



index5.html

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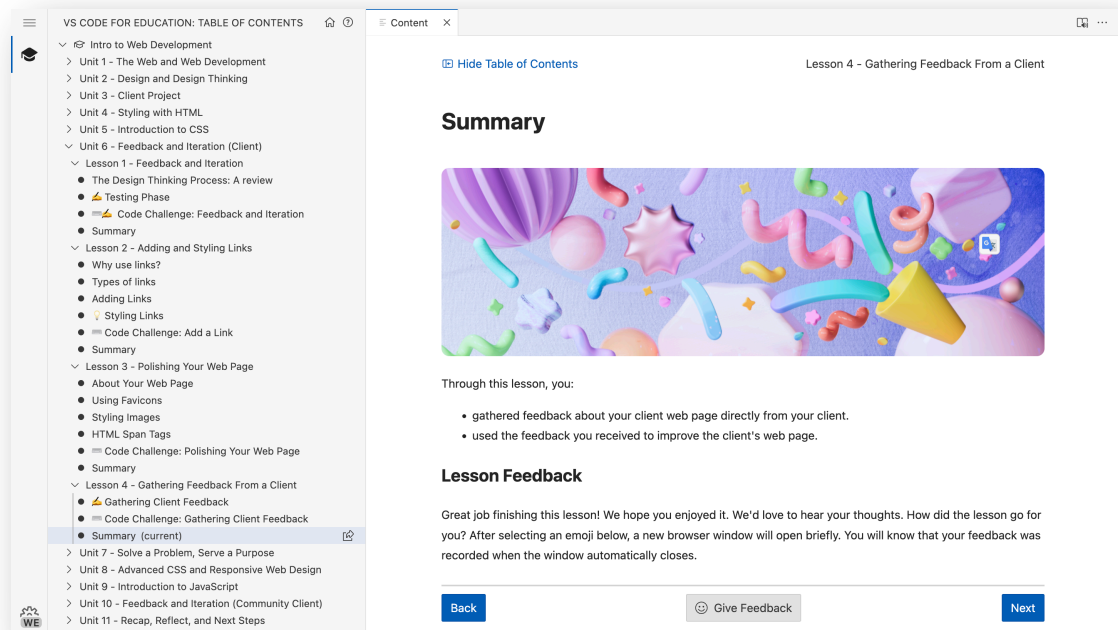
Part 3: Reflection

In Unit 5, I explored the essentials of CSS and its role in web design. Lesson 1 introduced CSS and its importance, where I practiced sizing an image using width and height properties in a code challenge, making my client's web page more visually appealing. Lesson 2 covered basic CSS properties like font-size, font-family, color, and background-color, which I used to style text and backgrounds effectively. Lesson 3 taught me the differences between inline, internal, and external CSS, and I successfully added an internal style sheet to my client's web page during a code challenge.

Initially, I found it tricky to decide which CSS method to use, but the hands-on exercises clarified their purposes—inline for quick fixes, internal for single pages, and external for larger projects. This unit showed me how CSS complements HTML to create engaging websites, laying a foundation for advanced styling in future lessons. I feel more confident in using CSS to enhance design and functionality now.

Unit 6 – Feedback and Iteration

Part 1: Screenshot of Work



Part 2: Code Snippet



index6.html



img

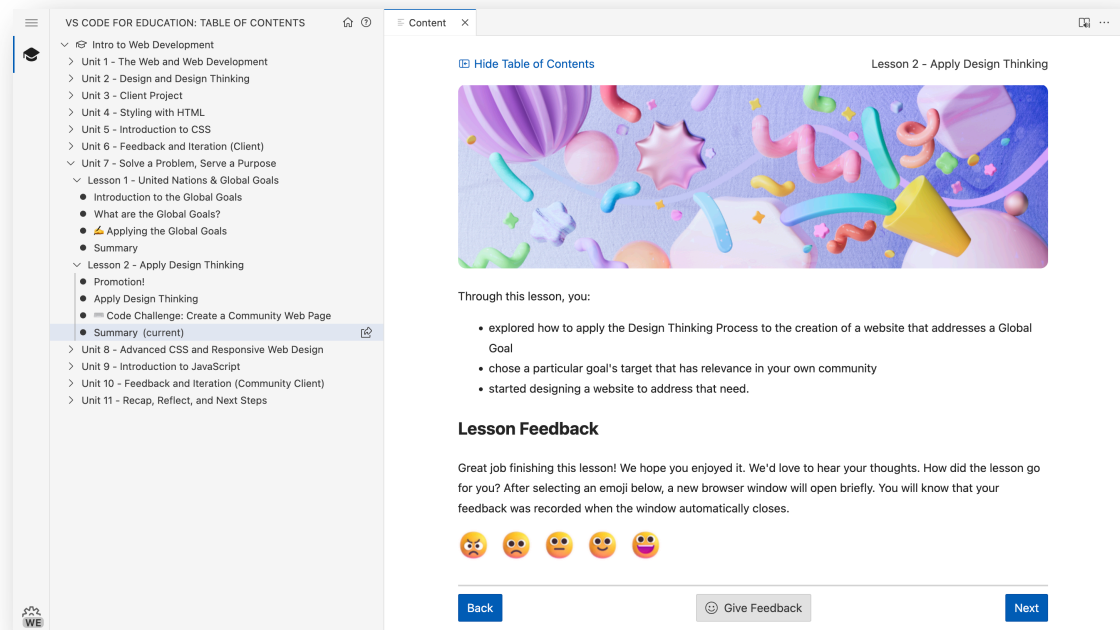
Uploaded in the zip file: unit_6.zip

Part 3: Reflection

Unit 6 taught me how to refine my web page through feedback and iteration. Lesson 1 emphasized the Design Thinking Process, where I gathered user feedback to improve readability and clarity. Lesson 2 showed me how to style links, enhancing the Spotify link's interactivity. Lesson 3 focused on polishing, so I added a Favicon, styled the image border with purple, and used `<span>` tags to highlight key text. Lesson 4 involved Shannon's feedback, leading me to add a tech logo and a placeholder button for interactivity. The biggest challenge was balancing feedback while maintaining Shannon's vision, which I overcame by prioritizing clarity and her purple theme. These changes improved the page's professionalism, accessibility, and engagement, making it more aligned with her tech focus. This unit highlighted the importance of iterative design in meeting client needs effectively.

Unit 7 – Solve a Problem, Serve a Purpose

Part 1: Screenshot of Work



Part 2: Code Snippet



index7.html



img

Uploaded in the zip file: unit_7.zip

Part 3: Reflection

Unit 7 challenged me to apply design thinking to a community project, transitioning from an individual webpage to a broader impact. In the Empathize stage, I interviewed a community member and identified the need for a website to promote environmental initiatives. The Define stage helped me frame the problem clearly, while Ideation allowed me to brainstorm multi-page designs with accessibility in mind. Prototyping was exciting—I created a basic webpage with styled images, links, and a green theme reflecting the environmental focus. The biggest challenge was ensuring the design was user-centered while maintaining simplicity, which I overcame by focusing on clear navigation and contrast. This unit taught me how to think globally and act locally, connecting local needs to global goals like SDG 13 (Climate Action). I feel more confident in designing for real-world impact and look forward to learning advanced CSS in Unit 8 to enhance my site further.

Unit 8 – Advanced CSS and Responsive Web Design

Part 1: Screenshot of Work

The screenshot displays the VS Code for Education interface. On the left, the 'TABLE OF CONTENTS' sidebar lists units from 1 to 11. Unit 8, 'Advanced CSS and Responsive Web Design', is expanded, showing Lesson 4, 'Responsive vs. Adaptive Design', as the current page. The main content area on the right shows the 'Summary' for Lesson 4, which includes a colorful abstract image and a list of learning objectives. At the bottom, there are 'Back', 'Give Feedback', and 'Next' buttons.

VS CODE FOR EDUCATION: TABLE OF CONTENTS

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 - Unit 1 - The Web and Web Development
 - Unit 2 - Design and Design Thinking
 - Unit 3 - Client Project
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 - Unit 5 - Introduction to CSS
 - Unit 6 - Feedback and Iteration (Client)
 - Unit 7 - Solve a Problem, Serve a Purpose
- Unit 8 - Advanced CSS and Responsive Web Design
 - Lesson 1 - Style Using div and class
 - Styles Using Div and Class
 - HTML Class Attribute
 - Code Challenge: Adding CSS to your Global Goals page
 - Summary
 - Lesson 2 - The Box Model and Positioning
 - The Box Model
 - CSS Position Property
 - Code Challenge: Box and Positioning
 - Summary
 - Lesson 3 - The Display Property and Positioning
 - What is the CSS display property?
 - Explore Values
 - Code Challenge: Display Property and Positioning
 - Summary
 - Lesson 4 - Responsive vs. Adaptive Design
 - Understanding Responsive versus Adaptive Design
 - Creating Responsive Design
 - Creating Adaptive Design
 - Choosing Responsive versus Adaptive Design
 - Code Challenge: Responsive and Adaptive Design
 - Summary (current)
 - Unit 9 - Introduction to JavaScript
 - Unit 10 - Feedback and Iteration (Community Client)
 - Unit 11 - Recap, Reflect, and Next Steps

Lesson 4 - Responsive vs. Adaptive Design

Summary

Through this lesson, you:

- were introduced to the fundamental concepts of responsive and adaptive design in web design
- learned how to code for each type of design and the importance of creating websites that work across multiple devices
- considered the advantages of each design and which design might be the most appropriate for your website

Lesson Feedback

Great job finishing this lesson! We hope you enjoyed it. We'd love to hear your thoughts. How did the lesson

[Back](#) [Give Feedback](#) [Next](#)

Part 2: Code Snippet



index8.html



img

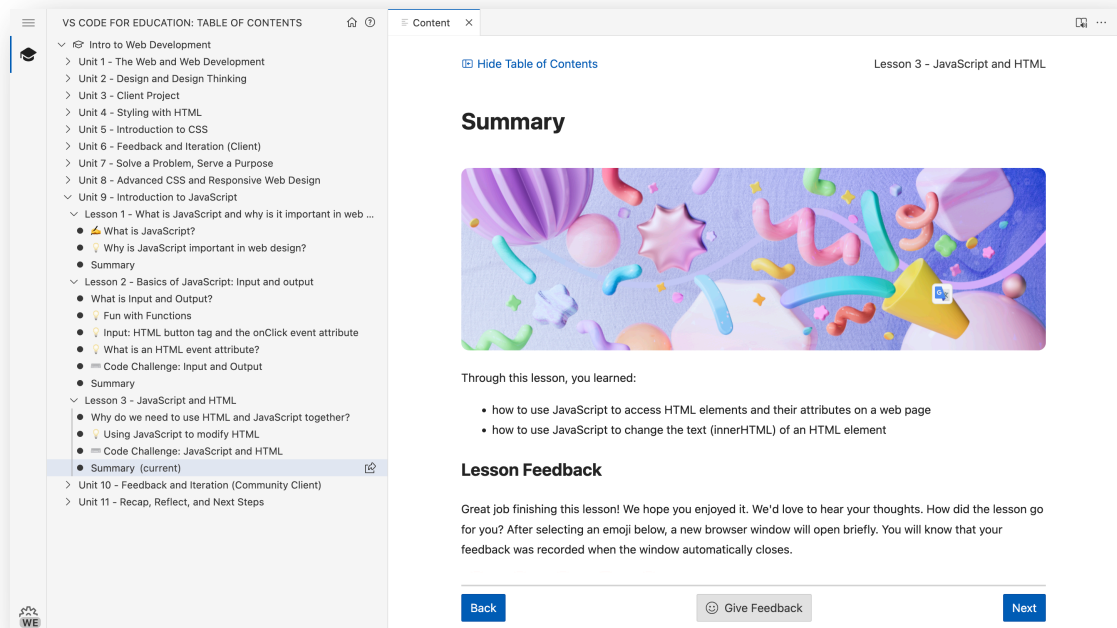
Uploaded in the zip file: unit_8.zip

Part 3: Reflection

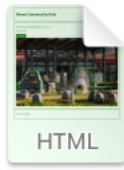
Unit 8 expanded my CSS skills with advanced layout techniques. Learning about `<div>` and class helped me organize content, while the box model and positioning taught me precise control over element spacing and placement. Exploring display values like flex and grid opened up new layout possibilities, making designs more dynamic. The responsive vs. adaptive design lesson was challenging yet rewarding—I created two versions of a community page, using fluid grids and media queries for responsiveness, and breakpoints for adaptability. The biggest hurdle was balancing flexibility with structure, which I overcame by testing layouts on different screen sizes. This unit deepened my understanding of user-friendly, accessible design, aligning with community needs like environmental awareness. I'm excited to apply these skills to multi-page sites and explore advanced CSS further!

Unit 9 – Introduction to JavaScript

Part 1: Screenshot of Work



Part 2: Code Snippet



index9.html



img



script.js

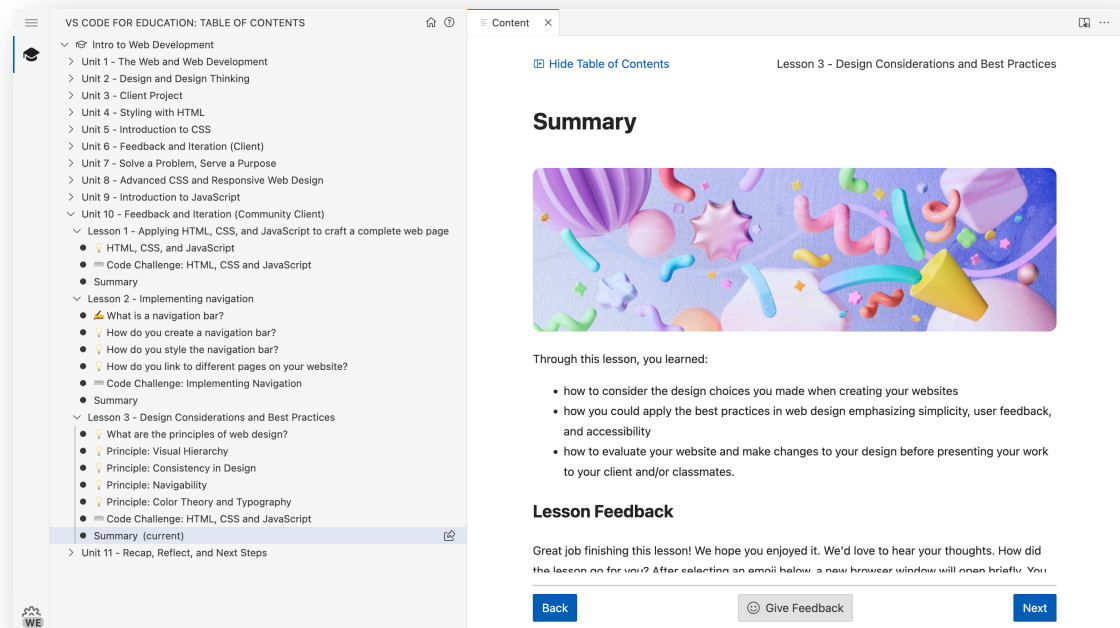
Uploaded in the zip file: unit_9.zip

Part 3: Reflection

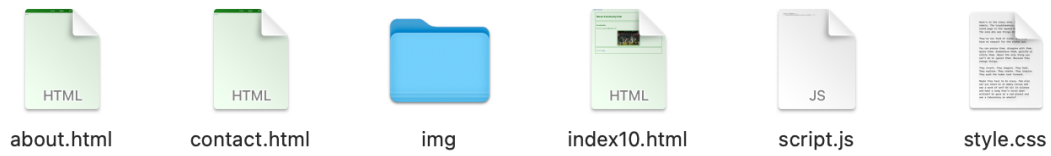
Unit 9 introduced me to JavaScript, enhancing my web design skills. Lesson 1 highlighted JavaScript's role in adding interactivity, which I applied by exploring sites like Spotify. Lesson 2 taught me input/output basics, using `alert()` and buttons, which I practiced with a pop-up and click event. Lesson 3 showed how JavaScript and HTML work together, inspiring me to create a dynamic text change on my Green Community Hub site. The challenge was ensuring the function worked seamlessly with HTML, which I overcame by separating the JavaScript into `script.js`. This unit deepened my understanding of dynamic websites, making the page more engaging. I enjoyed seeing my site evolve from static to interactive, and I'm excited to learn more JavaScript for complex features like forms or animations in future units!

Unit 10 – Feedback and Iteration

Part 1: Screenshot of Work



Part 2: Code Snippet



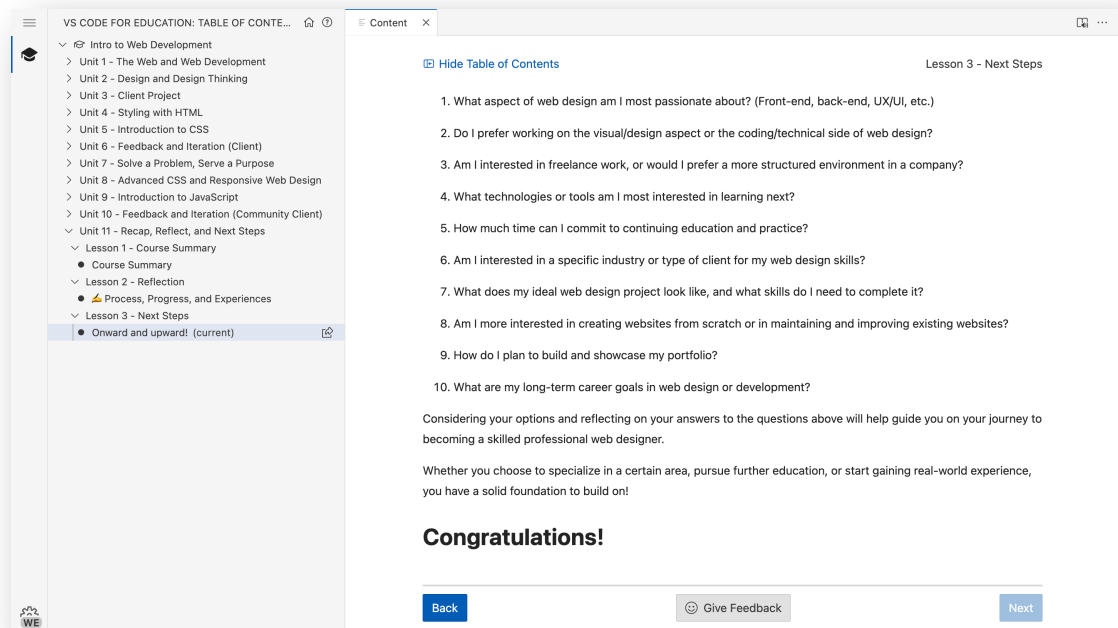
Uploaded in the zip file: unit_10.zip

Part 3: Reflection

Unit 10 deepened my understanding of web design through navigation and design principles. I implemented a navigation bar across `index.html`, `about.html`, and `contact.html`, applying HTML structure, CSS responsiveness with media queries, and JavaScript interactivity for a button function. Learning to style the navbar with gradients, shadows, and hover effects was challenging—I had to balance aesthetics with readability across devices. I overcame this by testing on mobile, tablet, and desktop, adjusting margins and breakpoints. Lessons on visual hierarchy and consistency helped me ensure the navbar enhances usability, while color theory guided my green theme. This unit highlighted navigation's role in user experience, making the site more intuitive. I'm proud of the cohesive multi-page design and excited to seek client feedback to iterate, perhaps adding dropdowns or icons, and explore advanced styling further!

Unit 11 – Recap, Reflect, and Next Steps

Part 1: Screenshot of Work



Part 2: Code Snippet

It is my FINAL PROJECT!

Part 3: Reflection

This web design course was a fun and exciting journey for me. In Unit 1, I started with HTML to make the basic parts of a webpage. I learned how to use tags like `<div>` to build a simple structure. In Units 6 and 7, I studied CSS to make my pages look nice. I learned about the box model, positioning, and display styles like flex and grid. I also used media queries to make my Green Community Hub look good on phones and tablets.

In Unit 8, I learned the difference between responsive and adaptive design. I made my website work well on different screens by using flexible images and grids. Unit 9 taught me JavaScript, starting with the `alert()` function to show messages. Then I added a button to change text when clicked. I put my JavaScript code in a separate file called `script.js` to keep things neat.

Unit 10 was the big finish. I made a website with three pages: `index.html`, `about.html`, and `contact.html`. I added a navigation bar to all pages and made it pretty with CSS. I used a green gradient background, a shadow, and hover effects. I also used JavaScript to highlight the current page link. The course taught me about accessibility, so I added alt text to images. I also thought about the UN Global Goals and used a green theme to support the environment.

I faced some hard parts, like fixing JavaScript errors and matching styles across files. But I kept trying and used W3Schools to help me. This course helped me grow a lot. I went from knowing nothing to making a full website that works well and looks good. I learned to think about users and make my site easy for everyone to use. I'm really proud of my progress. Now I want to learn more, like how to use React for better websites and study UI/UX to make designs even nicer. I also want to help my community by making websites for local groups to share ideas about the environment.